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EXPERTISE na-studio.pages.dev/expertise/finance-risk

PROFILE

Quantitative analyst with a dual background in applied mathematics and actuarial science, complemented by an MSc in Business Analytics. My insurance experience involved reserving, control and reporting data in a demanding environment. I combine model understanding, calculation tools and the ability to explain the assumptions, sensitivities and limitations behind an estimate.

Target positioning: reserving, risk, actuarial modelling, quantitative finance, time series and regulated reporting.

ACTUARIAL, FINANCE & RISK SKILLS

Actuarial & risk	Reserving, development triangles, Chain-Ladder, pricing, risk theory, Solvency II and IFRS 17.
Quantitative finance	Black-Scholes, derivatives, valuation, greeks, volatility scenarios, financial markets and pricing logic.
Statistics & modelling	Regression, time series, ARIMA / GARCH, stochastic models, classification, forecasting and validation.
Tools	R, SAS, Python, NumPy, SciPy, scikit-learn, Excel VBA, SQL, Jupyter, Tableau and Power BI.

PROFESSIONAL EXPERIENCE

Data Science & Actuarial Intern <i>SADA Assurances - Finance & Control Department</i> - Worked on technical reserving, consistency checks and development-triangle analysis with R, SAS and Excel. - Contributed to financial and actuarial reporting, calculation documentation and result interpretation.	Apr. - Aug. 2024
Data Analyst Intern <i>Iterato</i> - Analysed performance data, built indicators and produced structured reporting with Python, SQL and Excel. - Cleaned and strengthened data used for operational monitoring.	May - Jul. 2022
Operations Assistant & Quality Tracking <i>Dental Laboratory</i> - Tracked production information, quality controls and structured data in a technical environment. - Applied rigorous methods and maintained traceability of gaps.	Apr. - Aug. 2023
Administrative Agent - Tax Management & Control <i>Direction Générale des Finances Publiques (DGFIP)</i> - Processed and verified tax cases in a regulated and confidential setting. - Followed procedures, checked documents and maintained reliable information.	Summers 2016 · 2017 · 2018
Financial & Applied Mathematics Tutor <i>Universities & private tutoring</i> Taught analysis, probability, statistics and financial mathematics up to master level.	2017 - 2024

EDUCATION

MSc Business Analytics <i>Swiss Business School</i> Predictive modelling, quantitative analytics, data-driven decision making and applied visualisation.	2025 - 2026 · Ongoing
Master's in Banking, Finance & Actuarial Science <i>University of Montpellier</i> Actuarial science, risk, reserving, pricing, Solvency II, IFRS 17, quantitative finance, R, SAS and VBA.	2023 - 2024
Master's in Applied Mathematics for Engineering - MAPI3 <i>Toulouse III - Paul Sabatier University</i> Stochastic models, numerical analysis, optimisation, simulation and scientific programming.	2021 - 2023
Bachelor's in Mathematics <i>University of the Antilles</i> Analysis, probability, statistics, algebra and quantitative reasoning.	2017 - 2020

SELECTED FINANCE, ACTUARIAL & RISK PROJECTS

Machine Learning for Insurance Reserving Python / R · Chain-Ladder · Random Forest · Gradient Boosting · Validation Comparison of traditional actuarial methods and machine-learning approaches for technical reserve estimation. Work presented: data preparation, feature construction, estimate comparison, error analysis and interpretation of under- or over-reserving risk.	ACTUARIAL & ML
Black-Scholes Lab Python / R · Pricing · Greeks · Volatility · Scenarios Implementation and exploration of the Black-Scholes model for option valuation and sensitivity analysis. Work presented: theoretical pricing, greek analysis, volatility scenarios and visualisation of parameter effects.	QUANTITATIVE FINANCE
Brent 2023 - Econometric Study ARIMA / GARCH · Stationarity tests · Visualisation · Interpretation Study of Brent crude dynamics in 2023 to separate trend, volatility and isolated shocks. Work presented: series preparation, exploration, statistical testing, time-series modelling and synthesis with economic interpretation.	TIME SERIES
RL Crypto Trader Python · Reinforcement learning · Time series · Backtesting Experimental trading agent applying sequential decision logic to crypto-asset data. Work presented: environment, action and reward definition, series preparation, simulation and cautious result analysis.	FINANCE & AI

APPROACH

Assumptions → calculation → sensitivity → interpretation: check data quality, make assumptions explicit, compare models and scenarios, measure sensitivities and document limitations before drawing conclusions.
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LANGUAGES & ADDITIONAL INFORMATION

Languages	French: native · English: B2 · Spanish: B1.
Strengths & interests	Precision, reliability, critical thinking, teaching ability · Chess, fitness, reading, quantitative finance and digital creation.

Project details and sample deliverables are available through the [portfolio](#), [GitHub](#) and [LinkedIn](#).